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Studierichting: Informatica
Oefeningenreeks 6G nr. 2.8

$$f(x) = e^{|x|}$$

$$f(-x) = e^{|-x|} = e^{|x|} = f(x)$$

$$\Rightarrow f \text{ is even} \Rightarrow b_n = 0$$

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n \cos(nx))$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx = \frac{1}{\pi}\int_{-\pi}^{\pi} e^{|x|}dx$$

Omdat f even is:

$$= 2 \cdot \frac{1}{\pi}\int_0^{\pi} e^x dx$$

$$= \frac{2}{\pi}\left[e^x\right]_0^{\pi}$$

$$\Rightarrow a_0 = \frac{2}{\pi}(e^{\pi} - 1)$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)dx = \frac{1}{\pi}\int_{-\pi}^{\pi} e^{|x|}\cos(nx)dx$$

Omdat f even is:

$$= \frac{2}{\pi}\int_0^{\pi} e^x \cos(nx)dx$$

$$\left\{\begin{array}{l} u = \cos(nx) \\ dv = e^x dx \end{array}\right. \Rightarrow \left\{\begin{array}{l} du = -n\sin(nx)dx \\ v = e^x \end{array}\right.$$

$$I = e^x\cos(nx) - n\int_0^{\pi} e^x \sin(nx)dx$$

$$\left\{\begin{array}{l} u = \sin(nx) \\ dv = e^x dx \end{array}\right. \Rightarrow \left\{\begin{array}{l} du = n\cos(nx)dx \\ v = e^x \end{array}\right.$$

$$J = e^x\sin(nx) - n\int_0^{\pi} e^x \cos(nx)dx$$

$$\Rightarrow I = e^x\cos(nx) - n(e^x\sin(nx) - nI)$$

$$= e^x\cos(nx) - ne^x\sin(nx) + n^2 I$$

$$I(1+n^2) = e^x\cos(nx) - ne^x\sin(nx)$$

$$= e^x(\cos(nx) + n\sin(nx))$$

$$= \frac{2}{\pi}\left[\frac{e^x(\cos(nx) + n\sin(nx))}{1+n^2}\right]_0^\pi$$
$$= \frac{2}{\pi}\left(\frac{e^\pi(-1)^n}{1+n^2} - \frac{1}{1+n^2}\right)$$
$$\Rightarrow a_n = \frac{2(e^\pi(-1)^n - 1)}{\pi(1+n^2)}$$
$$f(x) \sim \frac{e^\pi - 1}{\pi} + \sum_{n=1}^{\infty} \frac{2(e^\pi(-1)^n - 1)}{\pi(1+n^2)} \cos(nx)$$

References